

Python Capstone Project 1

E-Commerce Order Analytics System

Business Scenario

An e-commerce company wants to analyze customer orders.

The company stores order data in CSV files and wants reports for:

```
Revenue Analysis
Product Analysis
Customer Analysis
City Analysis
Order Reports
```

Input File

orders.csv

```
order_id,customer_name,city,product,category,quantity,price
1001,Rahul Sharma,Hyderabad,Laptop,Electronics,1,65000
1002,Priya Reddy,Bangalore,Mobile,Electronics,2,25000
1003,Amit Kumar,Mumbai,Chair,Furniture,3,5000
1004,Sneha Patel,Chennai,Laptop,Electronics,1,70000
1005,Farhan Ali,Delhi,Table,Furniture,2,12000
1006,Neha Singh,Hyderabad,Mobile,Electronics,1,28000
1007,Arjun Verma,Pune,Chair,Furniture,4,5500
1008,Meera Nair,Bangalore,TV,Electronics,1,45000
1009,Kiran Rao,Hyderabad,Table,Furniture,1,13000
1010,Ravi Kumar,Mumbai,Mobile,Electronics,2,27000
1011,Deepak Jain,Delhi,Laptop,Electronics,1,68000
1012,Ananya Gupta,Hyderabad,TV,Electronics,2,47000
```

Part 1 – File Handling

Task 1

Read orders.csv.

Task 2

Display all records.

Task 3

Count total orders.

Part 2 – Revenue Analysis

Task 4

Calculate total revenue.

Formula:

```
Revenue = Quantity × Price
```

Task 5

Find highest order value.

Task 6

Find lowest order value.

Task 7

Find average order value.

Part 3 – Customer Analysis

Task 8

Display all unique customers.

Task 9

Count unique customers.

Task 10

Find customer with highest purchase amount.

Part 4 – Product Analysis

Task 11

Count orders by product.

Task 12

Calculate revenue by product.

Task 13

Find most sold product.

Task 14

Find least sold product.

Task 15

Calculate revenue by category.

Part 5 – City Analysis

Task 16

Count orders by city.

Task 17

Calculate revenue by city.

Task 18

Find city generating highest revenue.

Part 6 – Lists, Sets and Dictionaries

Task 19

Store all product names in a list.

Sort alphabetically.

Task 20

Store unique cities in a set.

Display all cities.

Task 21

Create dictionary:

```
{  
    city : revenue  
}
```

Task 22

Create dictionary:

```
{  
    product : quantity_sold  
}
```

Part 7 – Functions

Create functions:

Task 23

```
calculate_total_revenue()
```

Task 24

```
find_top_product()
```

Task 25

```
find_top_city()
```

Task 26

```
find_average_order_value()
```

Part 8 – Exception Handling

Task 27

Handle missing CSV file.

Task 28

Handle invalid quantity values.

Task 29

Handle invalid price values.

Part 9 – NumPy

Create NumPy array containing order values.

Task 30

Calculate:

```
Total Revenue  
Average Revenue  
Maximum Revenue  
Minimum Revenue  
Standard Deviation
```

Part 10 – Pandas

Task 31

Read CSV using Pandas.

Task 32

Create:

```
Revenue Column
```

Task 33

Display top 5 highest value orders.

Task 34

Group by city and calculate revenue.

Task 35

Group by product and calculate revenue.

Task 36

Display top selling products.

Task 37

Display city-wise order count.

Report Generation

Generate:

```
sales_summary_report.txt
```

Containing:

```
Total Orders
```

```
Total Revenue
```

```
Average Order Value
```

```
Highest Order Value
```

```
Lowest Order Value
```

```
Revenue By City
```

```
Revenue By Category
```

```
Top Selling Product
```

```
Top Revenue Generating City
```

Bonus Tasks

Task 38

Generate:

```
high_value_orders.csv
```

Orders above ₹50,000.

Task 39

Generate:

electronics_orders.csv

Only Electronics category.

Task 40

Create menu-driven application.

1. View Orders
2. Revenue Analysis
3. Product Analysis
4. City Analysis
5. Export Reports
6. Exit